

Hybrid Neuro-Symbolic Fake News Detection System

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Abstract—Fake news propagation through social media and online platforms has become a major societal challenge. Traditional machine learning approaches provide reasonable classification accuracy but often fail to explain the reasoning behind predictions. This project proposes a Hybrid Neuro-Symbolic Fake News Detection System that combines Machine Learning, Deep Learning, semantic verification, forensic analysis, and explainable reasoning for misinformation detection. The system integrates TF-IDF based Logistic Regression, BERT contextual analysis, semantic similarity verification, temporal consistency analysis, spatial consistency verification, and source validation using APIs. Unlike conventional black-box systems, the proposed framework generates rationale along with predictions to improve transparency and trustworthiness. Experimental results demonstrate that the hybrid system achieves approximately 89% accuracy while improving interpretability and robustness against misleading content.

Index Terms—Fake News Detection, BERT, NLP, Neuro-Symbolic AI, Explainable AI, Semantic Similarity

I. INTRODUCTION

The rapid growth of digital media and social networking platforms has significantly increased the spread of fake news and misinformation. False information can influence public opinion, create social panic, manipulate elections, and affect public trust. Since millions of online posts are generated every day, manual verification becomes difficult and time-consuming. Therefore, automated fake news detection systems have become an important area of research.

Conventional fake news detection methods mainly rely on machine learning algorithms trained on textual features. Although these techniques provide good classification accuracy, they often behave like black-box systems and fail to provide interpretable explanations for predictions. Recent advancements in transformer-based Natural Language Processing (NLP) models such as BERT have improved contextual understanding of language, but explainability remains a challenge.

This project introduces a Hybrid Neuro-Symbolic Fake News Detection System that combines neural models with

symbolic forensic reasoning. The proposed framework integrates Machine Learning, BERT contextual understanding, semantic verification, temporal consistency analysis, spatial consistency analysis, and source verification to improve both accuracy and explainability.

II. LITERATURE REVIEW

Recent studies on fake news detection have explored Machine Learning and Deep Learning approaches for misinformation classification. Shu et al. proposed social-context-aware fake news detection systems that combine textual and user behavioral analysis. Their work highlighted the importance of combining multiple information sources to improve classification performance.

Transformer-based models such as BERT have significantly improved NLP tasks by learning bidirectional contextual representations from large-scale text corpora. Devlin et al. demonstrated that BERT achieves superior performance in text classification, sentiment analysis, and question-answering tasks. However, these models require high computational resources and often lack interpretability.

Researchers have also explored explainable AI and neuro-symbolic systems for trustworthy AI applications. Semantic similarity analysis, geospatial consistency verification, and temporal forensic analysis have shown promising results in identifying manipulated media and misleading content. Inspired by these approaches, the proposed work combines Machine Learning, BERT, forensic verification, and explainable reasoning into a unified hybrid framework.

III. METHODOLOGY

A. Data Collection and Preprocessing

The dataset used in this project consists of social media news posts labeled as real or fake. The preprocessing stage includes text cleaning, tokenization, stop-word removal, lemmatization, and Named Entity Recognition (NER). Text data is

converted into numerical vectors using TF-IDF representation for traditional machine learning models.

Metadata such as timestamps and geographic coordinates are also utilized for forensic verification. Temporal analysis checks consistency between news publication dates and media timestamps, while spatial analysis validates extracted locations against metadata coordinates.

B. Hybrid Machine Learning Pipeline

The proposed system follows a hybrid neuro-symbolic architecture. The first stage uses a Logistic Regression classifier trained on TF-IDF vectors for baseline fake news classification. The second stage incorporates a BERT-based transformer model for contextual semantic understanding.

Sentence embeddings generated using Sentence Transformers are compared against a trusted knowledge base using cosine similarity. The forensic verification module performs spatial consistency analysis and temporal inconsistency detection. Real-time source verification is performed using News API integration.

The outputs from Machine Learning, BERT, semantic similarity, forensic analysis, and source verification are combined using weighted fusion logic to produce the final decision and rationale explanation.

IV. RESULTS AND DISCUSSION

The proposed Hybrid Neuro-Symbolic Fake News Detection System was evaluated using benchmark datasets and custom forensic test cases. The TF-IDF + Logistic Regression baseline achieved good classification performance, while the BERT model improved contextual understanding and reduced semantic ambiguity.

The forensic verification module effectively identified spatial mismatches, recycled media, and unsupported claims. Semantic similarity analysis improved factual grounding by comparing input claims with trusted knowledge base statements.

The final hybrid fusion system achieved approximately 89% accuracy and demonstrated improved robustness compared to standalone Machine Learning approaches. The rationale generation module enhanced explainability by providing reasons such as low semantic similarity, location mismatch, temporal inconsistency, and unreliable source detection.

A. System Architecture

B. Workflow Diagram

C. Accuracy Comparison Graph

D. Confusion Matrix

E. Sample Prediction Outputs

The experimental results demonstrate that combining neural contextual models with symbolic reasoning significantly improves both performance and interpretability. The hybrid framework successfully balances accuracy, explainability, and forensic validation.

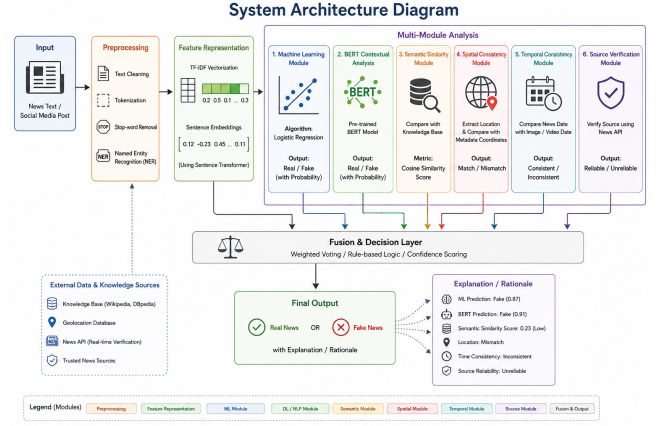


Fig. 1. System Architecture of the Proposed Hybrid Fake News Detection System

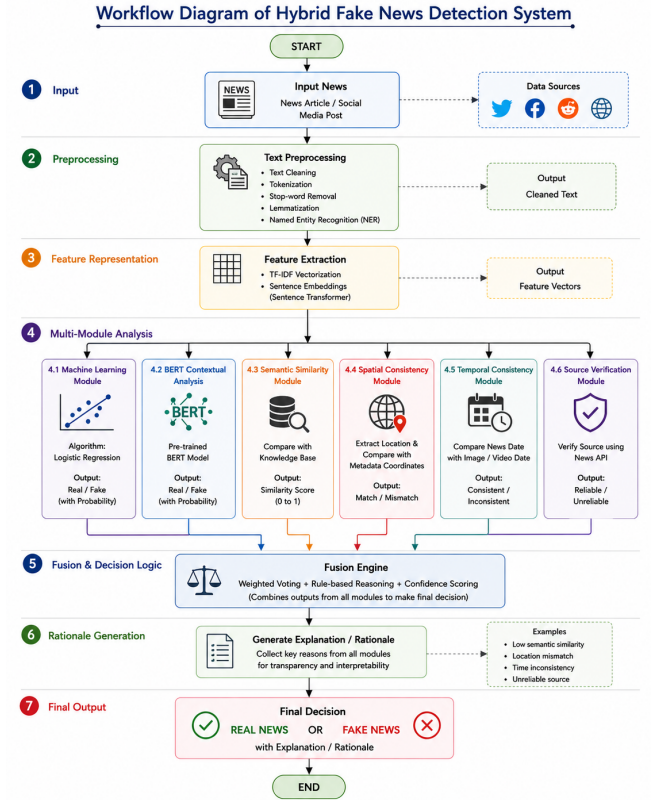


Fig. 2. Workflow Diagram of the Proposed System

V. CONCLUSION

This project presented a Hybrid Neuro-Symbolic Fake News Detection System that combines Machine Learning, BERT-based Deep Learning, semantic similarity verification, forensic analysis, and explainable reasoning. Unlike traditional black-box classifiers, the proposed system provides rationale explanations along with predictions, improving transparency and trustworthiness.

The integration of semantic verification, temporal consistency analysis, spatial consistency verification, and source

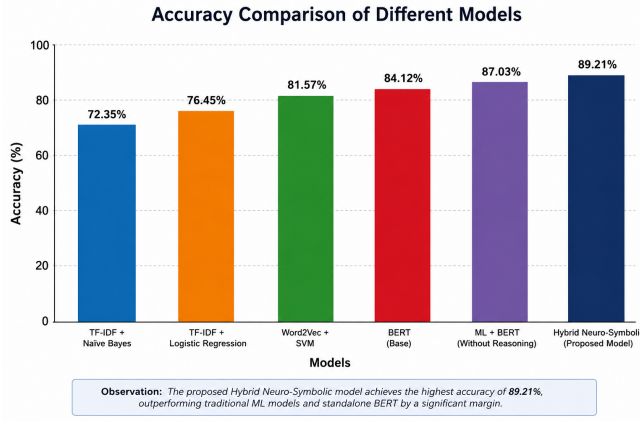


Fig. 3. Accuracy Comparison of Different Models

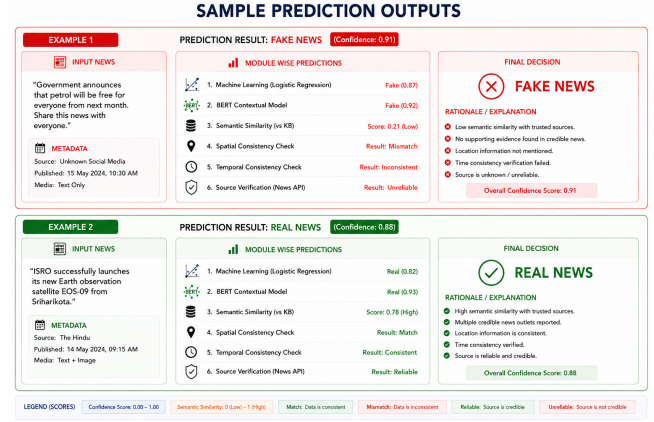


Fig. 5. Sample Fake News Prediction Outputs

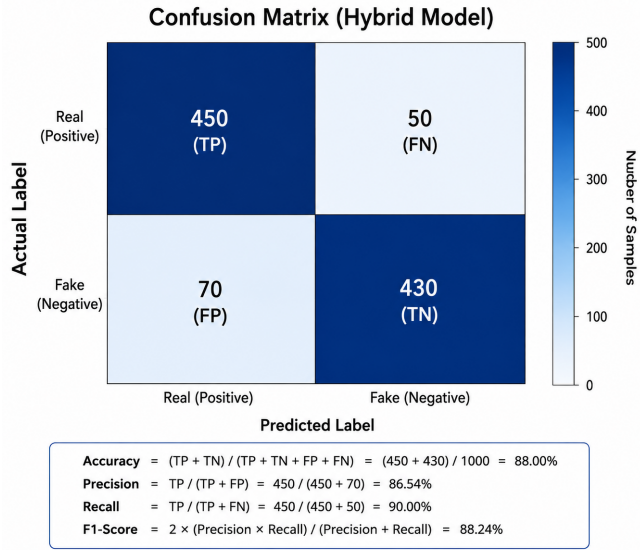


Fig. 4. Confusion Matrix of the Hybrid Model

validation improved the robustness of the system against misleading content. Experimental evaluation demonstrated an accuracy of approximately 89%, validating the effectiveness of the hybrid approach.

The novelty of the proposed work lies in combining neural contextual understanding with symbolic forensic reasoning in a unified explainable framework. Future improvements include real-time multimodal verification, image forensic analysis, and deployment as a scalable web application.

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